

Formal Axiomatic Monograph: Non-Standard Indeterminate

Logic $\mathcal{L}_\infty(A_0)$

Ultraproduct Model Semantics, Geodesic Kripke Structures, and Euler Phase Cancellation

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Abstract

We construct the formal non-standard logic system $\mathcal{L}_\infty(A_0)$ defined over an N -dimensional vector manifold anchored at the indeterminate zero-point seed $A_0 \equiv 0/0 = \lambda$. Using a non-principal ultrafilter $\mathcal{U}$ over an infinite index set, we construct the hyperreal valuation space ${}^*\mathbb{V} = \mathbb{V}^I/\mathcal{U}$ and establish a geodesic Kripke frame $\mathcal{K}_{\text{Lie}} = (W, \mathcal{R}, \mathcal{V})$ parameterized by the hyperfinite Lie group ${}^*SO(*N)$. We prove that the modal operators $\Box$ (Necessity) and $\Diamond$ (Possibility) map directly to continuous Lie algebra geodesics, satisfying S5 modal logic. Finally, we demonstrate that all self-referential paradoxes undergo destructive phase cancellation via Euler's Identity ($e^{i\pi} + 1 = 0$), forcing paradoxical amplitudes identically to zero ($|\psi_{\text{node}}| = 0$) and guaranteeing paradox-free model completeness.

1. Introduction and Primitive Alphabet

Classical binary and multi-valued logics operate over discrete or bounded intervals. In contrast, $\mathcal{L}_\infty(A_0)$ constructs a non-standard logical space where truth valuations reside on a continuous vector manifold $\mathbb{V} \subseteq \mathbb{R}^N(A_0)$.

Definition 1.1 (Primitive Alphabet Σ). *The logic system $\mathcal{L}_\infty(A_0)$ is defined over the alphabet $\Sigma = (\mathbb{V}, \mathcal{M}, \nabla_k, \otimes, \Phi_{\text{node}})$, where:*

- i. **Truth Manifold** ($\mathbb{V}$): Valuation domain anchored at the scalar seed $V_0 = 0/0 = \lambda$.*
- ii. **Mirror Operator** ($\mathcal{M}$): Inverts propositional scale across A_0 : $\mathcal{M}(P_N) = P_{-N} = \frac{1}{P_N}$.*
- iii. **Nodal Partition Operator** (∇_k): Generates $R_N = \binom{N}{2}$ orthogonal rotation planes.*
- iv. **Crystallization Operator** ($\otimes$): Fuses a proposition with its reciprocal mirror: $P_N \otimes \mathcal{M}(P_N) \equiv \mathcal{C}_N$.*
- v. **Kinetic Firewall** (Φ_{node}): Imposes zero-amplitude probability boundaries on paradoxes ($|\psi_{\text{node}}| = 0$).*

2. Axiomatic Foundation ($\mathcal{A}_{\text{non-std}}$)

Axiom 2.1 (Zero-Point Origin A_0). *Array index 0 occupies exclusive primacy, where the indeterminate seed $\lim_{x \rightarrow 0}(0/x) = \lambda \in \mathbb{R}^N$ generates coordinate-free propositional scale.*

Axiom 2.2 (Dimensional Mirror Symmetry). *Propositional capacity expands linearly in positive tiers ($P_N = 3 + N$) and inverts symmetrically into fractional anti-properties ($P_{-N} = \frac{1}{3+N}$) across A_0 .*

Axiom 2.3 (Tautological Crystallization). *The fusion of any property count with its fractional mirror inverse across A_0 yields a fixed physical invariant:*

$$P_N \otimes \mathcal{M}(P_N) = (3 + N) \otimes \left(\frac{1}{3 + N} \right) \equiv \mathcal{C}_N \quad (1)$$

Axiom 2.4 (Euler Phase Equilibrium). *For phase transformations $\theta \in [0, 2\pi]$ on $SO(2)$ rotation planes, half-period rotations ($\theta = \pi$) enforce anti-phase cancellation:*

$$e^{i\pi} + 1 = 0 \quad (2)$$

Axiom 2.5 (Nodal Firewall Isolation). *Contradictory and paradoxical state evaluations are restricted to zero-measure nodal surfaces:*

$$|\psi_{node}| = 0 \quad (3)$$

3. Ultraproduct Construction of ${}^*\mathbb{V}$

Let $I = \mathbb{N}$ be the index set of valuation tiers, and let $\mathcal{U} \subset \mathcal{P}(I)$ be a non-principal ultrafilter over I .

Definition 3.1 (Hyperreal Valuation Manifold). *The hyperreal truth manifold ${}^*\mathbb{V}$ is the quotient space:*

$${}^*\mathbb{V} = \left(\prod_{i \in I} \mathbb{V}_i \right) / \equiv_{\mathcal{U}} \quad (4)$$

where $\mathbf{f} \equiv_{\mathcal{U}} \mathbf{g} \iff \{i \in I \mid \mathbf{f}(i) = \mathbf{g}(i)\} \in \mathcal{U}$.

By Łoś's Theorem, every first-order formula valid in standard finite valuation tiers transfers strictly to ${}^*\mathbb{V}$. The indeterminate seed A_0 is represented by the infinitesimal halo $\mu(0) \subset {}^*\mathbb{V}$, containing hyperreal elements $\lambda^* = [(1, 1/2, 1/3, \dots)]_{\mathcal{U}}$.

4. Geodesic Kripke Semantics for Modal Operators

Definition 4.1 (Geodesic Kripke Frame). *We define the frame $\mathcal{K}_{Lie} = (W, \mathcal{R}, \mathcal{V})$ where:*

1. $W = \{w_{\mathbf{g}} = \mathbf{g} \cdot \mathbf{v}_0 \mid \mathbf{g} \in {}^*SO({}^*N)\}$.
2. $w_{\mathbf{g}_1} \mathcal{R} w_{\mathbf{g}_2} \iff \exists s \in {}^*\mathbb{R}, {}^*\mathbf{J} \in {}^*\mathfrak{so}({}^*N) \text{ s.t. } \mathbf{g}_2 = \mathbf{g}_1 \cdot \exp(s {}^*\mathbf{J})$.
3. $\mathcal{V}(\phi, w_{\mathbf{g}}) = 1 \iff \mathbf{g} \cdot \nu(\phi) \in \text{SAT}({}^*\mathbb{V})$.

Theorem 4.2 (S5 Modal Isomorphism). *The geodesic accessibility relation $\mathcal{R}$ generated by ${}^*SO({}^*N)$ defines an equivalence relation, establishing an exact model for S5 modal logic.*

Proof. a. **Reflexivity (T):** For $s = 0$, $\exp(0 \cdot {}^*\mathbf{J}) = \mathbf{I}$, so $w_{\mathbf{g}} \mathcal{R} w_{\mathbf{g}}$.

b. **Transitivity (4):** By the Baker-Campbell-Hausdorff formula, the product of two geodesic steps $\exp(s_1 {}^*\mathbf{J}_1) \cdot \exp(s_2 {}^*\mathbf{J}_2) = \exp(s_3 {}^*\mathbf{J}_3)$ remains inside ${}^*SO({}^*N)$.

c. **Symmetry (B):** For any generator ${}^*\mathbf{J}$, the inverse step $\exp(-s {}^*\mathbf{J}) = (\exp(s {}^*\mathbf{J}))^{-1}$ exists in ${}^*SO({}^*N)$.

Thus $\mathcal{R}$ is reflexive, transitive, and symmetric, fulfilling S5. $\square$

5. Euler Phase Cancellation and Paradox Elimination

Theorem 5.1 (Impossibility of Self-Referential Paradoxes). *In $\mathcal{K}_{Lie}$, any self-referential paradox ψ_{node} satisfies $\nu(\Diamond\psi_{node}) = 0$.*

Proof. Let ψ_{node} be a paradoxical formula whose truth value flips under a half-period rotation $\theta = \pi$ along a geodesic plane $^*\mathbf{J}$:

$$\nu(\psi_{node}(\mathbf{g} \cdot \exp(\pi^* \mathbf{J}))) = e^{i\pi} \cdot \nu(\psi_{node}(\mathbf{g})) = -\nu(\psi_{node}(\mathbf{g})) \quad (5)$$

Since a complete geodesic circuit requires $e^{i\pi} + 1 = 0$, we have:

$$\nu(\psi_{node}) = -\nu(\psi_{node}) \implies 2\nu(\psi_{node}) = 0 \implies \nu(\psi_{node}) = 0 \quad (6)$$

Thus, no world $w \in W$ satisfies ψ_{node} , proving $\nu(\Diamond\psi_{node}) = 0$. Self-referential contradictions have zero probability measure across all Lie geodesics. $\square$

Summary of Logical Completeness

The non-standard system $\mathcal{L}_\infty(A_0)$ resolves classical incompleteness by embedding valuation states into continuous Lie algebra geodesics. Paradoxes undergo destructive interference via $e^{i\pi} + 1 = 0$, rendering $\mathcal{L}_\infty(A_0)$ paradox-free and non-standardly complete.